

Advanced (Habanero)

- Q1.** What is the graph of the inequality $y > 2x - 3$?
 (A) Solid line with shading above
 (B) Dashed line with shading above
 (C) Solid line with shading below
 (D) Dashed line with shading below
 (E) No shading
- Q2.** Which of the following is the solution region for $x + y < 4$?
 (A) Triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 4)$
 (B) Triangle with vertices $(0, 0)$, $(0, 4)$, $(4, 0)$
 (C) Quadrant I with shading below the line $x + y = 4$
 (D) Quadrant I with shading above the line $x + y = 4$
 (E) Quadrant II with shading below the line $x + y = 4$
- Q3.** The inequality $y \geq -x + 2$ has a
 (A) Solid boundary line
 (B) Dashed boundary line
 (C) No boundary line
 (D) Vertical boundary line
 (E) Horizontal boundary line
- Q4.** What is the graph of $x - 2y > 5$?
 (A) Dashed line with shading below
 (B) Solid line with shading above
 (C) Dashed line with shading above
 (D) Solid line with shading below
 (E) No shading
- Q5.** The solution region for $y < 3x - 2$ is
- (A) Above the line $y = 3x - 2$
 (B) Below the line $y = 3x - 2$
 (C) On the line $y = 3x - 2$
 (D) To the left of the line $y = 3x - 2$
 (E) To the right of the line $y = 3x - 2$
- Q6.** Which inequality has the graph with a dashed boundary line and shading below?
 (A) $y > x - 1$
 (B) $y \geq x - 1$
 (C) $y < x - 1$
 (D) $y \leq x - 1$
 (E) $y = x - 1$
- Q7.** The inequality $x > 2$ has a
 (A) Vertical solid boundary line
 (B) Vertical dashed boundary line
 (C) Horizontal solid boundary line
 (D) Horizontal dashed boundary line
 (E) No boundary line
- Q8.** The solution region for the system $x + y > 2$ and $x - y < 1$ is
 (A) The intersection of two half-planes
 (B) The union of two half-planes
 (C) A single half-plane
 (D) A quadrant
 (E) An axis

Open Ended Questions (FRQ)

Q9. Graph the inequality $y > -2x + 1$ and identify the solution region.

Q10. Solve the system of inequalities $x + y < 3$ and $x - y > -2$. Graph the solution region and identify the vertices of the region.

Chef's Tips

- Tip 1: Use a graphing calculator to visualize the inequalities.
- Tip 2: Pay attention to the direction of the inequality when graphing.
- Tip 3: Check for any special cases, such as vertical or horizontal boundary lines.